

DIVYANSHU RAJ

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SUMMARY

Final-year CSE student at VIT-AP University with hands-on research experience in Deep Learning at IIT Patna. Skilled in Python, TensorFlow, and object detection systems, with a growing foundation in full stack development and cloud technologies. Certified in Oracle Generative AI, Neo4j, and MongoDB. Actively seeking SDE or ML Engineer roles where I can contribute to real-world products and continue growing as a developer.

EDUCATION

B.Tech in Computer Science & Engineering

2022 – Present

VIT-AP University, Amaravati

WORK EXPERIENCE

Research Intern — FIST-TBI, IIT Patna

Jun 2024 – Jul 2024 | On-site

- Collaborated with a faculty researcher on object detection using the YOLO deep learning architecture.
- Handled full ML pipeline: dataset preparation, model training, hyperparameter tuning, and evaluation.
- Gained hands-on exposure to a real-world academic R&D environment at a top-tier institution.

PROJECTS

Military Aircraft Detection | Python · TensorFlow · OpenCV · NumPy

- Designed and trained a custom CNN to classify military aircraft images, achieving 87% classification accuracy.
- Applied advanced data augmentation techniques to improve model generalization on a curated dataset.
- Optimized model architecture for lightweight deployment with robust bounding box localization.

GitHub: github.com/divyanshu1213/military-aircraft-detection

Object-Avoiding Car | Raspberry Pi Pico · MicroPython · Ultrasonic Sensors · LDR

- Built an autonomous prototype vehicle that detects and avoids obstacles in real time using embedded sensors.
- Programmed the Raspberry Pi Pico microcontroller to process sensor input and control motor responses.
- Integrated LDR light sensor for smarter environment-aware navigation alongside ultrasonic distance sensing.

GitHub: github.com/divyanshu1213/object-avoiding-car

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML, CSS, MySQL

Frameworks: TensorFlow, OpenCV, Streamlit

Tools: Git & GitHub, Google Colab, Postman, MongoDB Compass, Figma, AWS, Tableau

Interests: Generative AI, Deep Learning, Full Stack Development, Data Analysis

Soft Skills: Problem Solving, Leadership, Communication, Collaboration, Learning Agility

CERTIFICATIONS

- [Oracle Generative AI Professional](#)
- [Neo4j Certified Professional](#)
- [MongoDB Certified DBA](#)
- [Java 11 Essentials — Infosys Springboard](#)
- [Research Internship Certificate — FIST-TBI, IIT Patna](#)